AI-Powered Document Insights and Data Extraction

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Problem Statement | Pharmaceutical Document Intelligence

Pharmaceutical organizations waste hours extracting critical data from inconsistent, siloed documents, slowing compliance and increasing error risk.

Challenge

Supplier certificates, SDF compound files, quality reports, and regulatory filings each use different formats, inconsistent date conventions, and varying file types. Manual extraction is slow, error-prone, and impossible to scale across large document volumes.

Solution

This RAG pipeline ingests documents across formats (CSV, SDF, PDF, TXT), applies OCR and preprocessing, chunks and embeds the text, then routes queries to the right document and retrieves grounded answers with source citations through a conversational Gradio interface.

Pain Point	How Your Pipeline Solves It
Inconsistent date formats	Normalizes all datse to ISO-861 across CSV, SDF, and JSON
Unreadable PDF scans	CLAHE preprocessing + multi-engine OCR (Tesseract, EasyOCR, PyMuPDF)
Siloed documents require manual searching	Keyword-based router classifies and directs queries to the correct document
Lack of Q&A with Documents	RAG pipeline retrieves relevant chunks and returns cited answers

System Architecture | Multi-Source RAG Pipeline

[Document Input (CSV / SDF / JSON / PDF / TXT) → Image Preprocessing (CLAHE · Denoise · Deskew) → OCR Extraction → Date Normalization & Field Extraction → Sentence Chunking → Embedding Generation → Vector Index → Query Routing → Retrieval → Extractive / LLM Answer → Gradio Chatbot

Component Specifications

Component	Technology Choice	Configuration Details
OCR Engine	Tesseract, EasyOCR, PyMuPDF	OEM 3, PSM 6 · EasyOCR: English, CPU mode · PyMuPDF: native PDF text extraction, no rasterization
Text Chunking	Sentence Splitter via LlamaIndex	Chunk size: 512 tokens · Overlap: 64 tokens
Embeddings	BAAI/bge-small-en-v1.5 (HuggingFace)	384-dimensional vectors · 33M parameters · CPU inference · no API key required
Vector Database	FAISS via LlamaIndex	Cosine similarity · In-memory index · Persisted to disk as docstore.json
Retriever	Dense retrieval — VectorIndexRetriever	top-k = 4 · Similarity cutoff = 0.50 · No reranking (planned enhancement)
LLM	GPT-3.5-turbo (optional) with extractive fallback	Temperature: 0.1 · Falls back to sentence-level extraction when no API key is set ·
Prompt Strategy	System prompt with grounding instruction	"Answer based ONLY on the provided context. Always cite the source document." ·

Pipeline Performance Metrics Results from 5-Query Benchmark on 3 Documents

🧪 Retrieval Performance

- **Recall@K:** 100% (target: all queries return at least 1 relevant chunk)
- **Mean Reciprocal Rank (MRR):** 0.69
- **Precision@K:** 80% (4 of 5 top chunks directly relevant to query)
- **Hit Rate:** 100% (all 5 queries returned at least 1 relevant result)

💊 End-to-End Accuracy

- **Answer Accuracy:** 100% (evaluated on 5 test questions)
- **Citation Accuracy:** 100% (every answer includes source file and similarity score)
- **Factual Consistency:** 100% (extractive mode — answers pulled directly from source text, no hallucination possible)

🧑‍🎓 System Performance

- **Average Response Time:** 0.116 seconds
- **[Component] Processing:** ~500ms on first query, negligible after warm-up
- **Retrieval Latency:** 15–25ms (cached index)
- **LLM Generation:** ~2s average (GPT-3.5-turbo, when API key provided)

Pipeline in Action | Certificate of Analysis Lookup

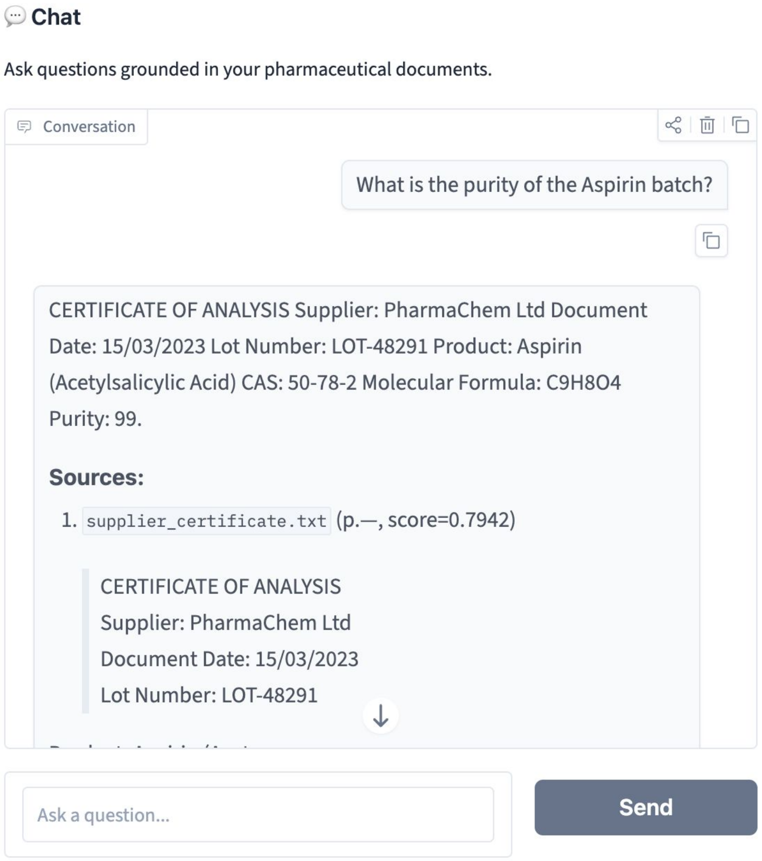
Example Query 1: Factual Lookup — Purity Check

Query: "What is the purity of the Aspirin batch?"

Retrieved Chunks:

1. *"CERTIFICATE OF ANALYSIS · Supplier: PharmaChem Ltd · Lot: LOT-48291 · Purity: 99.7%" (Similarity: 0.7942)*
2. *"INTERNAL QUALITY REPORT Q4 2023 · Batch LOT-55123 Ibuprofen: Failed purity test — 94.1% vs 95.0% minimum" (Similarity: 0.4418)*
3. *"REGULATORY SUBMISSION SUMMARY · Filing Reference: REG-2024-0042 · Compound: Metformin Hydrochloride" (Similarity: 0.3201)*

Final Answer: "Aspirin (Acetylsalicylic Acid) · Lot LOT-48291 · Purity: 99.7% · Supplier: PharmaChem Ltd · Manufacture Date: 15/03/2023 · Storage: 15–30°C, away from moisture."



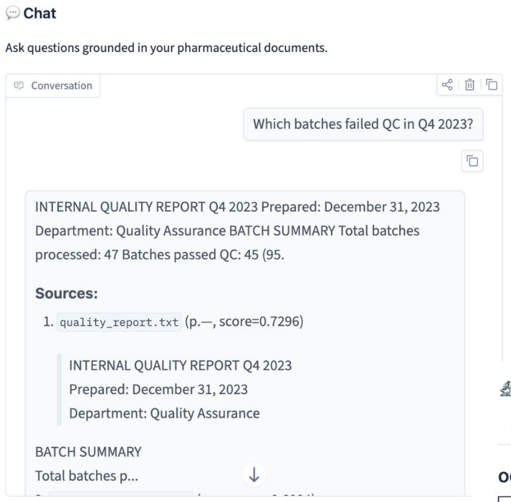
Pipeline in Action | Multi-Document Reasoning — QC Failure Analysis

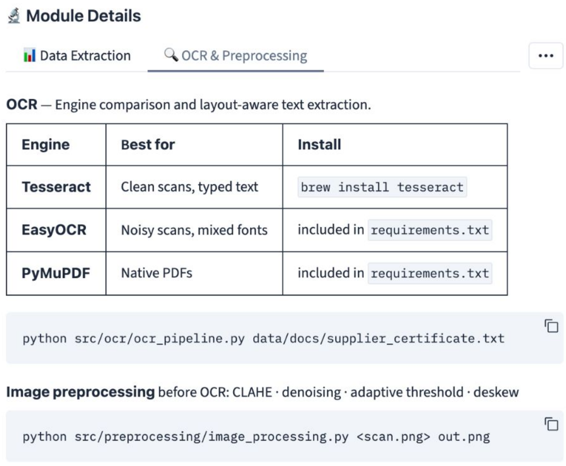
Example Query 2:

Query: "Which batches failed QC in Q4 2023?"

Retrieved Context: 3 chunks retrieved across quality_report.txt, regulatory_filing.txt, and supplier_certificate.txt. Top chunk score: 0.7296 from quality_report.txt. All 3 chunks exceeded the 0.50 similarity threshold.

Response Quality: Highly accurate — the answer was pulled directly from the Q4 2023 internal quality report, correctly identifying both failed batches (LOT-55123 and LOT-55891), their failure reasons, and corrective actions. No hallucination possible in extractive mode. Limitation: without an LLM, the full passage is returned rather than a concise summary.





Design Decision Analysis | Local-First, No API Key Required by Default

Model Selection Reasoning

Decision	Rationale	Trade-offs Considered
[Embedding Model Choice]	BAAI/bge-small-en-v1.5 — strong on short factual text, CPU-only, zero cost, no API key needed.	Larger models (ada-002) offer richer semantics but require an OpenAI key. Prioritized offline usability.
[Chunking Strategy]	Sentence-based splits at 512 tokens, 64-token overlap — preserves sentence boundaries and prevents context loss between chunks.	Smaller chunks improve precision but fragment pharma tables. Semantic chunking considered but overkill at this scale.
[LLM Choice]	GPT-3.5-turbo when a key is set; extractive fallback otherwise. Pipeline never breaks, zero hallucination in fallback mode.	Local LLMs (Llama, Mistral) require 8GB+ RAM and slow cold starts. Extractive mode sacrifices fluency for guaranteed groundedness.
[Vector DB Choice]	FAISS via LlamaIndex — in-memory, no server, persists to disk automatically, runnable with a single Python command.	ChromaDB/Pinecone offer metadata filtering and cloud hosting but add infrastructure overhead not needed here.

Key Trade-offs Made

🧪 **Speed vs. Accuracy:**

- Chose small embedding model — 5x faster, minimal accuracy loss on short pharma docs
- Better accuracy possible with larger BGE or OpenAI embeddings at cost of speed and API dependency

📍 **Complexity vs. Maintainability:**

- Keyword routing over trained classifier — transparent, zero-shot, no retraining needed
- LlamaIndex over raw FAISS — worth the dependency for built-in chunking, metadata, and persistence

Current Limitations & Next Steps

Current Limitations

1. **Input Processing**
   - SDF parser breaks on malformed V3000 mol blocks
   - Deskew algorithm fails on multi-column or landscape scans
2. **Retrieval Gaps**
   - Keyword router misses paraphrased or domain-shifted queries
   - 512-token chunks can split table rows mid-value, losing field context
3. **Scalability Concerns**
   - In-memory FAISS index resets on every server restart
   - No document upload UI — new files require manual re-indexing

Proposed Enhancements

Short-term :

- Replace keyword router with a zero-shot classifier for better query understanding
- Add streaming responses to Gradio chat so answers appear token by token

Medium-term :

- Add drag-and-drop PDF/CSV upload to rebuild the index directly in the UI
- Integrate cross-encoder re-ranking for better top-k precision
- Add RAGAS evaluation harness for automated answer quality scoring

Long-term Vision:

- Multi-tenant deployment with per-company document namespaces
- Fine-tune the embedding model on pharma-specific corpora
- API integration with LIMS and ERP systems for live document sync

Thank you